

SERIES FORMS OF EXACT DENSITIES ASSOCIATED WITH UNIFORMLY DISTRIBUTED RANDOM POINTS IN AN n -BALL

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Summary: This paper considers the r -content of the r -simplex whose $r+1$ vertices are uniformly and independently distributed random points of which p are in the interior and $r+1-p$ are on the boundary of a unit n -ball. Different computable representations of the exact density of the r -content in the general case are given by establishing a connection with some distribution problems in multivariate analysis.

1. INTRODUCTION

Consider a set of $r+1$ random points which are independently and uniformly distributed of which p are in the interior and $q=r+1-p$ are on the boundary of a unit n -ball, with $1 \leq r \leq n$, $p \geq 0$, $q \leq r+1$. Let Δ_n denote $r!$ times the r -content of the r -simplex generated by these points. Hammersley (1950) has given the exact density of Δ_n for $r=1$, $p=2$, $q=0$, Miles (1971) that for $r=1$, $p=2$, $q=0$; $r=1$, $p=1$, $q=1$; $r=1$, $p=0$, $q=2$. Mathai (1982) has obtained the exact density for $r=2$, $p=3$, $q=0$, while

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Mathai and Tracy (1983) that for $r=2, p=2; r=2, p=1; r=2, p=0$. Miles (1971) had conjectured that $(2n/r)^{\frac{1}{2}}[\Delta_n - (r+1)^{\frac{1}{2}}]$ is asymptotically standard normal as $n \rightarrow \infty$, and later Ruben (1977) proved that the conjecture was correct. A simpler proof for this conjecture is given in Mathai (1982). Moreover, in Mathai (1983) it is shown that similar results hold true for other types of r -contents. In particular it is pointed out that the exact density, in the general case, can be written in terms of G - and H -functions and therefore one can use various representations of these functions in order to write the density in convenient forms.

The aim of the present article is to give some explicit series forms for the density function in the general case so that computations can be carried out by using these representations. These computable representations are achieved by establishing a connection with respect to distribution problems in multivariate statistical analysis and then applying some of the techniques available there.

2. GENERAL MOMENTS

The h -th moment of Δ_n^2 is given (see Mathai and Tracy (1983)) by the following equation

$$(2.1) \quad E(\Delta_n^2)^h = \frac{\Gamma[(r+1)\frac{n}{2} + (r+1)h - r + p] \left\{ \prod_{j=0}^{r-1} \Gamma(h + \frac{n-j}{2}) \right\} [\Gamma(1 + \frac{n}{2})]^p [\Gamma(\frac{n}{2})]^{r+1-p}}{\Gamma[(r+1)\frac{n}{2} + rh - r + p] \left\{ \prod_{j=0}^{r-1} \Gamma(h + \frac{n-j}{2}) \right\} [\Gamma(h + 1 + \frac{n}{2})]^p [\Gamma(h + \frac{n}{2})]^{r+1-p}}$$

where, as usual, E denotes mathematical expectation. Note that (2.1) is valid for complex h defined in a certain strip as well, and therefore we can employ the theory of the inverse Mellin transform in order to derive the density of Δ_n^2 . However we will not use this technique in such a context because, as pointed out in Mathai (1982), it leads to expressions for the density in terms of G - and H -functions.

For convenience we begin by rewriting the gammas in (2.1) in the following form:

$$(2.2) \quad \Gamma(mz) = (2\pi)^{\frac{1-m}{2}} m^{\frac{mz-1}{2}} \prod_{j=0}^{m-1} \Gamma(z + \frac{j}{m}), \quad m = 1, 2, \dots$$

Let $U = \frac{r^r}{(r+1)^{r+1}} \Delta_n^2$. Then $E(U^h)$ with the help of (2.1) and (2.2) can be written in the following way:

$$(2.3) \quad E(U^h) = C \frac{\prod_{j=0}^r \Gamma[h + \frac{n}{2} - \frac{(r-p)}{r+1} + \frac{j}{r+1}] \prod_{j=1}^r \Gamma[h + \frac{n}{2} - \frac{j-1}{2}]}{\left\{ \prod_{j=0}^{r-1} \Gamma[h + \frac{n}{2}(\frac{r+1}{r}) - \frac{(r-p)}{r} + \frac{j}{r}] \right\} [\Gamma(h + 1 + \frac{n}{2})]^p [\Gamma(h + \frac{n}{2})]^{r-p+1}}$$

where

$$(2.4) \quad C = \frac{(2\pi)^{-\frac{1}{2}} (r+1)^{\frac{(r+1)n}{2} - r + p - \frac{1}{2}} [\Gamma(1 + \frac{n}{2})]^p [\Gamma(\frac{n}{2})]^{r-p+1}}{r^{\frac{(r+1)n}{2} - r + p - \frac{1}{2}} \prod_{j=1}^r \Gamma(\frac{n}{2} - \frac{j-1}{2})}$$

Here (2.3) is obtained by expanding the gammas $\Gamma[(r+1)\frac{n}{2} + (r+1)h - r + p]$ and $\Gamma[(r+1)\frac{n}{2} + rh - r + p]$ by taking $m = (r+1)$, r and $z = h + \frac{n}{2} - \frac{(r-p)}{(r+1)}$, $h + \frac{n}{2} \frac{(r+1)}{r} - \frac{(r-p)}{r}$ respectively and then using (2.2).

Now, by comparing the moment expression in (2.3) to that coming from the product of independent beta type-1 random variables we can easily see that U can be looked upon as a product of independent beta type-1 variables (see Mathai (1983)). A real random variable X_j is said to have a beta type-1 density if the density function is given by

$$(2.5) \quad f_j(x) = \frac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j)\Gamma(\beta_j)} x^{\alpha_j-1} (1-x)^{\beta_j-1}, \quad 0 < x < 1, \quad \alpha_j > 0, \quad \beta_j > 0$$

and $f_j(x) = 0$ otherwise. In this case the h -th moment of X_j is given by

$$(2.6) \quad E(X_j^h) = \frac{\Gamma(\alpha_j + h)}{\Gamma(\alpha_j)} \frac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j + \beta_j + h)}.$$

Let $Y = X_1 X_2 \dots X_p$ where $X_1, X_2, \dots, X_p$ are independent beta type-1 random variables having the densities as in (2.5). Then the h -th moment of Y is given by

$$(2.7) \quad E(Y^h) = \prod_{j=1}^p E(X_j^h) = C_p \prod_{j=1}^p \frac{\Gamma(\alpha_j + h)}{\Gamma(\alpha_j + \beta_j + h)}$$

with $C_p = \prod_{j=1}^p \frac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j)}$. On comparing (2.7) with (2.3) we realize that

U in (2.3) can be considered to be a product of $p = 2r+1$ independent beta type-1 real random variables with the following parameters:

$$(2.8) \quad \alpha_j = \frac{n}{2} - \frac{j-1}{2}, \quad j = 1, \dots, r$$

$$= \frac{n}{2} - \frac{(r-p)}{r+1} + \frac{(j-r-1)}{r+1}, \quad j = r+1, \dots, 2r+1$$

and

$$\beta_j = \frac{j-1}{2}, \quad j = 1, \dots, r-p+1$$

$$= 1 + \frac{j-1}{2}, \quad j = r-p+2, \dots, r$$

$$= 1 + \frac{(r-p)}{r+1}, \quad j = r+1$$

$$= \frac{n}{2r} - \frac{(r-p)}{r(r+1)} - \frac{1}{r} + \frac{(j-r-1)}{r(r+1)}, \quad j = r+2, \dots, 2r+1.$$

Note that this is only one way of writing the parameter pairs (α_j, β_j) . They can be paired in any convenient form provided $\alpha_j > 0$, $\beta_j > 0$ for all j . If in the notation of (2.8) some of the α_j 's and β_j 's are zero, then it will be seen that the corresponding gammas get cancelled.

3. EXACT DENSITY IN COMPUTABLE FORMS

The random variable U in (2.3) can be looked upon as a product of independent beta type-1 random variables with the parameters given in (2.8).

Let us derive the density of U by the technique of transformation of variables. Wilks (1932) established the following result for the density $g(u)$ of $U = X_1 X_2 \dots X_p$ where $X_1, \dots, X_p$ are independent and distributed according to (2.5).

$$(3.1) \quad g(u) = D_p u^{\alpha_p - 1} (1-u)^{\beta_1 + \dots + \beta_p - 1} \int_0^1 \dots \int_0^1 \prod_{j=1}^{p-1} v_j^{\beta_j - 1} (1-v_j)^{\beta_p + \beta_{p-1} + \dots + \beta_{p-j+1} - 1} \xi_j^{\alpha_j - (\alpha_{j+1} + \beta_{j+1})} dv_j$$

where

$$(3.2) \quad \xi_j = [v_1 + v_2(1-v_1) + \dots + v_j(1-v_1)(1-v_2) \dots (1-v_{j-1})](1-u)$$

and

$$(3.3) \quad D_p = \prod_{j=1}^p \frac{\Gamma(\alpha_j + \beta_j)}{\Gamma(\alpha_j)\Gamma(\beta_j)}.$$

Tretter and Walster (1975) noted that the ξ_j of (3.2) satisfy the recurrence relation $\xi_j = v_j + (1-v_j)\xi_{j-1}$, $j = 1, \dots, p-1$. By successive multiplication and integration the same authors reduced the density $g(u)$ to the following form:

$$(3.4) \quad g(u) = D_p u^a (1-u)^b \sum_j \sigma_j (1-u)^j$$

where a , b and σ_j do not contain u . This is one representation of the density function in computable form. Unfortunately, this representation has the disadvantage that σ_j becomes too complicated for programming purposes, and therefore we shall consider an alternative procedure.

By using the technique of transformation of variables and taking $U_1 = X_1 X_2$, $U_2 = X_1 X_2 X_3 = U_1 X_3, \dots, U = X_1 X_2 \dots X_p$, we can easily arrive at the following representation for the density function of U .

$$(3.5) \quad g(u) = D_p u^{\alpha_p - 1} (1-u)^{\delta_p - 1} \int_0^1 \dots \int_0^1 \prod_{j=1}^{p-1} t_j^{\delta_j - 1} (1-t_j)^{\beta_{j+1} - 1} [1-t_{p-1}t_{p-2} \dots t_{p-j}(1-u)]^{-\gamma_{p-j}} dt_j, \quad 0 < u < 1$$

with D_p as defined in (3.3), $\delta_j = \beta_1 + \beta_2 + \dots + \beta_j$, $\gamma_j = \alpha_{j+1} + \beta_{j+1} - \alpha_j$,

$$j = 1, \dots, p-1.$$

Note that $0 < t_{p-1}t_{p-2}\dots t_{p-j}(1-u) < 1$ and hence $[1 - t_{p-1}t_{p-2}\dots t_{p-j}(1-u)]^{-\gamma_{p-j}}$ can be expanded by using a binomial expression, that is,

$$(3.6) \quad [1 - t_{p-1}t_{p-2}\dots t_1(1-u)]^{-\gamma_1} = \sum_{k_1=0}^{\infty} \frac{(\gamma_1)_{k_1}}{k_1!} t_{p-1}^{k_1} \dots t_1^{k_1} (1-u)^{k_1}$$

$$[1 - t_{p-1}\dots t_2(1-u)]^{-\gamma_2} = \sum_{k_2=0}^{\infty} \frac{(\gamma_2)_{k_2}}{k_2!} t_{p-1}^{k_2} \dots t_2^{k_2} (1-u)^{k_2}$$

⋮

$$[1 - t_{p-1}(1-u)]^{-\gamma_{p-1}} = \sum_{k_{p-1}=0}^{\infty} \frac{(\gamma_{p-1})_{k_{p-1}}}{k_{p-1}!} t_{p-1}^{k_{p-1}} (1-u)^{k_{p-1}}$$

where, for instance, $(a)_m = a(a+1)\dots(a+m-1)$. Next, on multiplying and collecting the terms containing t_j , one has the following,

$$(3.7) \quad \int_0^1 t_j^{k_1+k_2+\dots+k_j+\delta_j-1} (1-t_j)^{\beta_{j+1}-1} dt_j = \frac{\Gamma(\delta_j+K_j)\Gamma(\beta_{j+1})}{\Gamma(\delta_{j+1}+K_j)}$$

where $K_j = k_1+k_2+\dots+k_j$, and δ_j is defined as before. Now, by multiplying and integrating all the t_j 's one derives the following result:

$$(3.8) \quad g(u) = D_p \left[\prod_{j=2}^p \Gamma(\beta_j) \right] u^{\alpha_p-1} (1-u)^{\delta_{p-1}} \sum_{k=0}^{\infty} \sum_K \prod_{j=1}^{p-1} \frac{k_j}{k_j!} \frac{\Gamma(\delta_j+K_j)}{\Gamma(\delta_{j+1}+K_j)} (1-u)^k$$

for $0 < u < 1$ and $g(u) = 0$ elsewhere, where K denotes a partitioning of the non-negative integer k into $p-1$ parts $K = (k_1, k_2, \dots, k_{p-1})$ such that $k_j \geq 0$, $k_1 + \dots + k_{p-1} = k$, $\sum_K$ being the summation over this partition.

To summarize we state the following:

Theorem:

Let $U = r^r \Delta_n^2 / (r+1)^{r+1}$ where Δ_n is $r!$ times the r -content of the r -simplex whose $r+1$ vertices are independently and uniformly distributed random points of which p are in the interior and $r+1-p$ are on the boundary of a unit n -ball. Then the exact density of U , for general values of r , n and p , is given by $g(u)$ of (3.8) where the parameters α_j , $j=0,1,\dots,2r-1$ and β_j , $j=0,1,\dots,2r-1$ are given in (2.8).

Note that (3.8) is in a form which is suitable for computation: the multiple series component is in fact already in a form convenient for programming purposes. From this expression it can be seen that the cumulative probabilities can be evaluated by means of incomplete beta tables. When using the series form for $g(u)$ it is advisable to cancel all possible gammas in $E(U^h)$ and identify p in the series (3.8) which is in fact the resulting number of gamma ratios after cancellation of the common gammas. Then (α_j, β_j) can be selected in the most suitable way.

4. A SPECIAL CASE

For the sake of illustration we will present the particular case where $r=1$, $p=2$, $q=0$. In this case the expression (2.3) becomes

$$(4.1) \quad E(U^h) = \frac{2^n [\Gamma(1 + \frac{n}{2})]^2}{\Gamma(\frac{1}{2})\Gamma(\frac{n}{2})} \frac{\Gamma(h + \frac{n}{2} + \frac{1}{2}) \Gamma(h + \frac{n}{2})}{\Gamma(h + n + 1)\Gamma(h + \frac{n}{2} + 1)}$$

$E(U^0) = 1$, as can be seen by expanding $\Gamma(n+1) = \Gamma[2(\frac{n}{2} + \frac{1}{2})]$. By comparing (4.1) with (2.7) we realize that

$$(4.2) \quad \alpha_1 = \frac{n}{2} + \frac{1}{2}, \quad \alpha_2 = \frac{n}{2} \quad \text{and} \quad \beta_1 = \frac{n}{2} + \frac{1}{2}, \quad \beta_2 = 1.$$

From (4.2) and knowing that

$$\delta_j = \beta_1 + \beta_2 + \dots + \beta_j \quad \text{and} \quad \gamma_j = \alpha_{j+1} + \beta_{j+1} - \alpha_j$$

we obtain

$$\gamma_1 = \alpha_2 + \beta_2 - \alpha_1 = \frac{1}{2} \quad \text{and} \quad \delta_1 = \frac{n}{2} + \frac{1}{2}, \quad \delta_2 = \frac{n}{2} + \frac{3}{2}.$$

From (3.8) we have

$$\begin{aligned} (4.3) \quad g(u) &= \frac{\Gamma(\alpha_1 + \beta_1)\Gamma(\alpha_2 + \beta_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\beta_1)} u^{\alpha_2-1} (1-u)^{\frac{n}{2} + \frac{1}{2}} \sum_{k=0}^{\infty} \frac{(\frac{1}{2})_k}{k!} \frac{\Gamma(\frac{n}{2} + \frac{1}{2} + k)}{\Gamma(\frac{n}{2} + \frac{3}{2} + k)} (1-u)^k \\ &= \frac{\Gamma(n+1)\Gamma(\frac{n}{2} + 1)}{\Gamma(\frac{n}{2} + \frac{1}{2})\Gamma(\frac{n}{2})\Gamma(\frac{n}{2} + \frac{1}{2})} u^{\frac{n}{2}-1} (1-u)^{\frac{n}{2} + \frac{1}{2}} \frac{\Gamma(\frac{n}{2} + \frac{1}{2})}{\Gamma(\frac{n}{2} + \frac{3}{2})} \sum_{k=0}^{\infty} \frac{(\frac{1}{2})_k (\frac{n}{2} + \frac{1}{2})_k}{k! (\frac{n}{2} + \frac{3}{2})_k} (1-u)^k \end{aligned}$$

by using $\Gamma(a+k) = \Gamma(a)(a)_k$. It follows that we can write

$$(4.4) \quad g(u) = \frac{\Gamma(n+1)\Gamma(\frac{n}{2} + 1)}{\Gamma(\frac{n}{2})\Gamma(\frac{n}{2} + \frac{1}{2})\Gamma(\frac{n}{2} + \frac{3}{2})} u^{\frac{n}{2}-1} (1-u)^{\frac{n}{2} + \frac{3}{2} - 1} F\left(\frac{1}{2}, \frac{n}{2} + \frac{1}{2}; \frac{n}{2} + \frac{3}{2}; 1-u\right)$$

where $F(\cdot)$ is a hypergeometric function. From the duplication formula for gamma functions one has the following relations:

$$\begin{aligned} \Gamma(n+1) &= n\Gamma(n) = n\Gamma\left[2\left(\frac{n}{2}\right)\right] = n(2\pi)^{-\frac{1}{2}} 2^{\frac{n}{2}-\frac{1}{2}} \Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{n}{2} + \frac{1}{2}\right) \\ &= \frac{n2^{n-1}\Gamma(\frac{n}{2})\Gamma(\frac{n}{2} + \frac{1}{2})}{\Gamma(\frac{1}{2})} \end{aligned}$$

and hence the expression (4.4) reduces to the form

$$(4.5) \quad g(u) = \frac{n2^{n-1}\Gamma(\frac{n}{2} + 1)}{\Gamma(\frac{1}{2})\Gamma(\frac{n}{2} + \frac{3}{2})} u^{\frac{n}{2}-1} (1-u)^{\frac{n}{2} + \frac{3}{2} - 1} F\left(\frac{1}{2}, \frac{n}{2} + \frac{1}{2}; \frac{n}{2} + \frac{3}{2}; 1-u\right)$$

with $0 < u < 1$. Let $u = \frac{\Delta^2}{4}$ and denote the density of Δ by $f(\Delta)$. Then

$$(4.6) \quad f(\Delta) = \frac{n\Gamma(\frac{n}{2} + 1)}{\Gamma(\frac{1}{2})\Gamma(\frac{n}{2} + \frac{3}{2})} \Delta^{n-1} \left(1 - \frac{\Delta^2}{4}\right)^{\frac{n}{2} + \frac{3}{2} - 1} F\left(\frac{1}{2}, \frac{n}{2} + \frac{1}{2}; \frac{n}{2} + \frac{3}{2}; 1 - \frac{\Delta^2}{4}\right)$$

with $0 < \Delta < 1$. However, the density of Δ provided by Miles (1971) for this particular case reads as

$$(4.7) \quad f(\Delta) = n\Delta^{n-1} \frac{\Gamma(1+\frac{n}{2})}{\Gamma(\frac{1}{2})\Gamma(\frac{n}{2}+\frac{1}{2})} \int_0^{1-(\Delta/2)^2} t^{(n-1)/2} (1-t)^{-\frac{1}{2}} dt.$$

It is not difficult to show that (4.6) and (4.7) are one and the same. In fact, consider the integral

$$(4.8) \quad \int_0^a t^\alpha (1-t)^{-\beta} dt = a^{\alpha+1} \int_0^1 w^\alpha [1-aw]^{-\beta} dw \\ = a^{\alpha+1} \frac{\Gamma(\alpha+1)\Gamma(1)}{\Gamma(\alpha+2)} F(\beta, \alpha+1; \alpha+2; a)$$

and then substitute $\alpha = \frac{n-1}{2}$, $\beta = \frac{1}{2}$, $a = 1 - \frac{\Delta^2}{4}$.

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